$$\frac{\partial Q}{\partial \beta_0} = 0 \Rightarrow \sum_{i=1}^{n} \{y_i - \beta_0 - \beta_1 x_i\} = 0$$

$$\Rightarrow \sum_{i=1}^{n} y_i - n\beta_0 - \beta_1 \sum_{i=1}^{n} x_i = 0 \quad \text{——— (1)}$$

$$\frac{\partial Q}{\partial \beta_1} = 0 \Rightarrow \sum_{i=1}^{n} x_i \{y_i - \beta_0 - \beta_1 x_i\} = 0$$

$$\Rightarrow \sum_{i=1}^{n} x_i y_i - \beta_0 \sum_{i=1}^{n} x_i - \beta_1 \sum_{i=1}^{n} x_i^2 = 0 \quad \text{——— (2)}$$

$$n\beta_0 + \beta_1 \sum_{i=1}^{n} x_i - \sum_{i=1}^{n} y_i = 0$$

$$\beta_0 \Sigma x_i + \beta_1 \Sigma x_i^2 - \Sigma x_i y_i = 0$$

$$\beta_0 + \beta_1 \frac{1}{n} \Sigma x_i - \frac{1}{n} \Sigma y_i = 0$$

or, $\beta_0 + \beta_1 \bar{x} - \bar{y} = 0$

or, $\boxed{\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}}$

$$\frac{\beta_0}{\;} = \frac{\beta_1}{n\Sigma x_i y_i - \Sigma x_i \Sigma y_i} = \frac{1}{n\Sigma x_i^2 - (\Sigma x_i)^2}$$

$$\Rightarrow \beta_1 = \frac{n\Sigma x_i y_i - \Sigma x_i \Sigma y_i}{n\Sigma x_i^2 - (\Sigma x_i)^2}$$

$$\bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x} \quad \text{--- (3)}$$

$$= \frac{n\Sigma x_i y_i - (n\bar{x})(n\bar{y})}{n\Sigma x_i^2 - n^2 \bar{x}^2}$$

$$\boxed{\hat{\beta}_1 = \frac{\Sigma x_i y_i - n\bar{x}\bar{y}}{\Sigma x_i^2 - n\bar{x}^2}} = \frac{\Sigma(x_i - \bar{x})(y_i - \bar{y})}{\Sigma(x_i - \bar{x})^2}$$

$$\boxed{y = \hat{\beta}_0 + \hat{\beta}_1 x} \quad \text{——— (4)}$$

[

$$y = \beta_0 + \beta_1 x + \beta_2 x^2$$

$$Q(\beta_0, \beta_1, \beta_2) = \sum_{i=1}^{n} \{y_i - (\beta_0 + \beta_1 x_i + \beta_2 x_i^2)\}^2$$]

④ − ③ : The regression line of y on x is

$$\boxed{y - \bar{y} = \hat{\beta}_1 (x - \bar{x})},$$ where

$$\hat{\beta}_1 = \frac{\sum x_i y_i - n\bar{x}\bar{y}}{\sum x_i^2 - n\bar{x}^2}$$

$$= \frac{\sum x_i y_i - n\bar{x}\bar{y}}{\sqrt{\sum x_i^2 - n\bar{x}^2}\sqrt{\sum y_i^2 - n\bar{y}^2}} \cdot \frac{\sqrt{\sum y_i^2 - n\bar{y}^2}}{\sqrt{\sum x_i^2 - n\bar{x}^2}}$$

or, $$\boxed{\hat{\beta}_1 = r \frac{\sigma_y}{\sigma_x} = b_{yx}}$$ (say)

→ Regression coefficient of y on x.

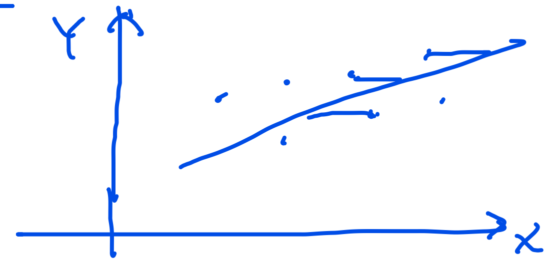
$$\sigma_y^2 = \frac{1}{n}\sum_{i=1}^{n}(y_i - \bar{y})^2, \quad \sigma_x^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

$$\boxed{y - \bar{y} = r\frac{\sigma_y}{\sigma_x}(x - \bar{x})}$$

$y = a + bx$

Line of regression of x on y:

$$y = \beta_0 + \beta_1 x$$



(x_i, y_i)

$$x - \bar{x} = b_{xy}(y - \bar{y})$$

where b_{xy} is the regression coefficient of x on y and is given by $b_{xy} = \dfrac{\Sigma x_i y_i - n\bar{x}\,\bar{y}}{\Sigma y_i^2 - n\bar{y}^2}$

$$\text{or, } \boxed{b_{xy} = r\,\frac{\sigma_x}{\sigma_y}}$$

$$\boxed{x - \bar{x} = r\,\frac{\sigma_x}{\sigma_y}(y - \bar{y})}$$

Properties: $b_{xy} \cdot b_{yx} = r\,\dfrac{\sigma_x}{\sigma_y} \cdot r\,\dfrac{\sigma_y}{\sigma_x}$

$$\therefore\ b_{xy} \cdot b_{yx} = r^2 \leq 1$$

$$\text{or, } \pm\sqrt{b_{xy} \cdot b_{yx}} = r$$

$\Rightarrow$ Both the regression coefficients must have the same algebraic sign.

Ex: For the variables x and y, the equations of the two regression lines are $4x+y=52$ and $x+y=32$. What is the correlation coefficient between x and y?

Soln: $4x+y=52$ or $x=-\frac{1}{4}y+13$

(as the line of regression of x on y) $b_{xy}=-\frac{1}{4}$

$x+y=32$ or $y=-x+32$

as the line of regression of y on x

$$b_{yx}=-1$$

$b_{xy}\cdot b_{yx}=\frac{1}{4}<1$

$\therefore \quad r^2=\frac{1}{4}$

$\Rightarrow r=-\frac{1}{2}$

$b_{xy}=r\left(\frac{\sigma_x}{\sigma_y}\right)$

$y=-4x+52$ $\quad$ $b_{yx}=-4$

$x=-y+32$ $\quad$ $b_{xy}=-1$

$\left.\begin{matrix} b_{yx}=-4 \\ b_{xy}=-1 \end{matrix}\right\}$ $b_{yx}\,b_{xy}=4>1$

Ex: Find the most likely price in Bombay corresponding to the price of Rs. 70 at Kolkata from the following data

	Kolkata	Bombay
Average Price	65	67
S. D.	2.5	3.5

The correlation coefficient between the prices of commodities in the two cities is 0.8.

Soln: Let the price of a commodity in Kolkata be x and that of a commodity in Bombay be y.

The regression line of y on x is

$$y - \bar{y} = r\frac{s_y}{s_x}(x - \bar{x});$$

$\bar{y} = 67$, $\bar{x} = 65$

$s_y = 3.5$, $s_x = 2.5$

$r = 0.8$

or, $y - 67 = 0.8 \times \frac{3.5}{2.5}(x - 65)$

or, $y - 67 = 1.12x - 72.8$

So the most likely price in Bombay corresponding to the price of Rs. 70 at Kolkata is

Rs $(67 + 1.12 \times 70 - 72.8)$ = Rs 72.60.